

14 Interrupts and events

14.1 Nested vectored interrupt controller (NVIC)

14.1.1 NVIC main features

- 74 maskable interrupt channels (not including the sixteen Cortex-M4 with FPU interrupt lines)
- 16 programmable priority levels (4 bits of interrupt priority are used)
- Low-latency exception and interrupt handling
- Power management control
- Implementation of System Control Registers

The NVIC and the processor core interface are closely coupled, which enables low latency interrupt processing and efficient processing of late arriving interrupts.

All interrupts including the core exceptions are managed by the NVIC. For more information on exceptions and NVIC programming, refer to the PM0214 programming manual for Cortex-M4 products.

14.1.2 SysTick calibration value register

The SysTick calibration value is set to 9000, which gives a reference time base of 1 ms with the SysTick clock set to 9 MHz ($\max f_{HCLK}/8$).

14.1.3 Interrupt and exception vectors

[Table 82](#) is the vector table for STM32F303xB/C and STM32F358xC devices. [Table 83](#) is the vector table for STM32F303x6/8 and STM32F328x8 devices.

Table 82. STM32F303xB/C/D/E, STM32F358xC and STM32F398xE vector table

Position	Priority	Type of priority	Acronym	Description	Address
-	-	-	-	Reserved	0x0000 0000
-	-3	Fixed	Reset	Reset	0x0000 0004
-	-2	Fixed	NMI	Non maskable interrupt. The RCC clock security system (CSS) is linked to the NMI vector.	0x0000 0008
-	-1	Fixed	HardFault	All classes of fault	0x0000 000C
-	0	Settable	MemManage	Memory management	0x0000 0010
-	1	Settable	BusFault	Pre-fetch fault, memory access fault	0x0000 0014
-	2	Settable	UsageFault	Undefined instruction or illegal state	0x0000 0018
-	-	-	-	Reserved	0x0000 001C - 0x0000 0028

Table 82. STM32F303xB/C/D/E, STM32F358xC and STM32F398xE vector table (continued)

Position	Priority	Type of priority	Acronym	Description	Address
-	3	Settable	SVCall	System service call via SWI instruction	0x0000 002C
-	5	Settable	PendSV	Pendable request for system service	0x0000 0038
-	6	Settable	SysTick	System tick timer	0x0000 003C
0	7	Settable	WWDG	Window watchdog interrupt	0x0000 0040
1	8	Settable	PVD	PVD through EXTI Line16 detection interrupt	0x0000 0044
2	9	Settable	TAMPER_STAMP	Tamper and TimeStamp interrupts through EXTI Line19	0x0000 0048
3	10	Settable	RTC_WKUP	RTC wakeup timer interrupt through EXTI Line20	0x0000 004C
4	11	Settable	FLASH	Flash global interrupt	0x0000 0050
5	12	Settable	RCC	RCC global interrupt	0x0000 0054
6	13	Settable	EXTI0	EXTI Line0 interrupt	0x0000 0058
7	14	Settable	EXTI1	EXTI Line1 interrupt	0x0000 005C
8	15	Settable	EXTI2_TS	EXTI Line2 and Touch sensing interrupts	0x0000 0060
9	16	Settable	EXTI3	EXTI Line3	0x0000 0064
10	17	Settable	EXTI4	EXTI Line4	0x0000 0068
11	18	Settable	DMA1_Channel1	DMA1 channel 1 interrupt	0x0000 006C
12	19	Settable	DMA1_Channel2	DMA1 channel 2 interrupt	0x0000 0070
13	20	Settable	DMA1_Channel3	DMA1 channel 3 interrupt	0x0000 0074
14	21	Settable	DMA1_Channel4	DMA1 channel 4 interrupt	0x0000 0078
15	22	Settable	DMA1_Channel5	DMA1 channel 5 interrupt	0x0000 007C
16	23	Settable	DMA1_Channel6	DMA1 channel 6 interrupt	0x0000 0080
17	24	Settable	DMA1_Channel7	DMA1 channel 7 interrupt	0x0000 0084
18	25	Settable	ADC1_2	ADC1 and ADC2 global interrupt	0x0000 0088
19 ⁽¹⁾	26	Settable	USB_HP/CAN_TX	USB high priority/CAN_TX interrupts	0x0000 008C
20 ⁽¹⁾	27	Settable	USB_LP/CAN_RX0	USB low priority/CAN_RX0 interrupts	0x0000 0090
21	28	Settable	CAN_RX1	CAN_RX1 interrupt	0x0000 0094
22	29	Settable	CAN_SCE	CAN_SCE interrupt	0x0000 0098
23	30	Settable	EXTI9_5	EXTI Line[9:5] interrupts	0x0000 009C
24	31	Settable	TIM1_BRK/TIM15	TIM1 break/TIM15 global interrupts	0x0000 00A0
25	32	Settable	TIM1_UP/TIM16	TIM1 update/TIM16 global interrupts	0x0000 00A4
26	33	Settable	TIM1_TRG_COM /TIM17	TIM1 trigger and commutation/TIM17 interrupts	0x0000 00A8
27	34	Settable	TIM1_CC	TIM1 capture compare interrupt	0x0000 00AC

Table 82. STM32F303xB/C/D/E, STM32F358xC and STM32F398xE vector table (continued)

Position	Priority	Type of priority	Acronym	Description	Address
28	35	Settable	TIM2	TIM2 global interrupt	0x0000 00B0
29	36	Settable	TIM3	TIM3 global interrupt	0x0000 00B4
30	37	Settable	TIM4	TIM4 global interrupt	0x0000 00B8
31	38	Settable	I2C1_EV	I2C1 event interrupt & EXTI Line23 interrupt	0x0000 00BC
32	39	Settable	I2C1_ER	I2C1 error interrupt	0x0000 00C0
33	40	Settable	I2C2_EV	I2C2 event interrupt & EXTI Line24 interrupt	0x0000 00C4
34	41	Settable	I2C2_ER	I2C2 error interrupt	0x0000 00C8
35	42	Settable	SPI1	SPI1 global interrupt	0x0000 00CC
36	43	Settable	SPI2	SPI2 global interrupt	0x0000 00D0
37	44	Settable	USART1	USART1 global interrupt & EXTI Line 25	0x0000 00D4
38	45	Settable	USART2	USART2 global interrupt & EXTI Line 26	0x0000 00D8
39	46	Settable	USART3	USART3 global interrupt & EXTI Line 28	0x0000 00DC
40	47	Settable	EXTI15_10	EXTI Line[15:10] interrupts	0x0000 00E0
41	48	Settable	RTC_Alarm	RTC alarm interrupt	0x0000 00E4
42 ⁽¹⁾	49	Settable	USBWakeUp	USB wakeup from Suspend (EXTI line 18)	0x0000 00E8
43	50	settable	TIM8_BRK	TIM8 break interrupt	0x0000 00EC
44	51	settable	TIM8_UP	TIM8 update interrupt	0x0000 00F0
45	52	settable	TIM8_TRG_COM	TIM8 Trigger and commutation interrupts	0x0000 00F4
46	53	settable	TIM8_CC	TIM8 capture compare interrupt	0x0000 00F8
47	54	settable	ADC3	ADC3 global interrupt	0x0000 00FC
48	55	Settable	FMC ⁽²⁾	FMC global interrupt	0x0000 0100
49	56	-	Reserved		0x0000 0104
50	57	-	Reserved		0x0000 0108
51	58	Settable	SPI3	SPI3 global interrupt	0x0000 010C
52	59	Settable	UART4	UART4 global and EXTI Line 34 interrupts	0x0000 0110
53	60	Settable	UART5	UART5 global and EXTI Line 35 interrupts	0x0000 0114
54	61	Settable	TIM6_DAC	TIM6 global and DAC1 underrun interrupts.	0x0000 0118
55	62	settable	TIM7	TIM7 global interrupt	0x0000 011C
56	63	Settable	DMA2_Channel1	DMA2 channel1 global interrupt	0x0000 0120
57	64	Settable	DMA2_Channel2	DMA2 channel2 global interrupt	0x0000 0124
58	65	Settable	DMA2_Channel3	DMA2 channel3 global interrupt	0x0000 0128
59	66	Settable	DMA2_Channel4	DMA2 channel4 global interrupt	0x0000 012C
60	67	Settable	DMA2_Channel5	DMA2 channel5 global interrupt	0x0000 0130

Table 82. STM32F303xB/C/D/E, STM32F358xC and STM32F398xE vector table (continued)

Position	Priority	Type of priority	Acronym	Description	Address
61	68	settable	ADC4	ADC4 global interrupt	0x0000 0134
62	69	-	Reserved		0x0000 0138
63	70	-	Reserved		0x0000 013C
64	71	Settable	COMP1_2_3	COMP1 & COMP2 & COMP3 interrupts combined with EXTI Lines 21, 22 and 29 interrupts.	0x0000 0140
65	72	Settable	COMP4_5_6	COMP4 & COMP5 & COMP6 interrupts combined with EXTI Lines 30, 31 and 32 interrupts.	0x0000 0144
66	73	settable	COMP7	COMP7 interrupt combined with EXTI Line 33 interrupt	0x0000 0148
67	74	-	Reserved		0x0000 014C
68	75	-	Reserved		0x0000 0150
69	76	-	Reserved		0x0000 0154
70	77	-	Reserved		0x0000 0158
71	78	-	Reserved		0x0000 015C
72	79	Settable	I2C3_EV ⁽²⁾	I2C3 event interrupt	0x0000 0160
73	80	Settable	I2C3_ER ⁽²⁾	I2C3 Error interrupt	0x0000 0164
74	81	Settable	USB_HP	USB High priority interrupt	0x0000 0168
75	82	Settable	USB_LP	USB Low priority interrupt	0x0000 016C
76	83	Settable	USB_WakeUp_RMP (see note 1)	USB wake up from Suspend and EXTI Line 18	0x0000 0170
77	84	settable	TIM20_BRK ⁽²⁾	TIM20 Break interrupt	0x0000 0174
78	85	settable	TIM20_UP ⁽²⁾	TIM20 Upgrade interrupt	0x0000 0178
79	86	settable	TIM20_TRG_COM ⁽²⁾	TIM20 Trigger and Commutation interrupt	0x0000 017C
80	87	settable	TIM20_CC ⁽²⁾	TIM20 Capture Compare interrupt	0x0000 0180
81	88	Settable	FPU	Floating point interrupt	0x0000 0184
82	89	-	-	Reserved	0x0000 0188
83	90	-	-	Reserved	0x0000 018C
84	91	Settable	-	SPI4 SPI4 global interrupt ⁽²⁾	0x0000 0190

1. It is possible to remap the USB interrupts (USB_HP, USB_LP and USB_WKUP) on interrupt lines 74, 75 and 76 respectively by setting the USB_IT_RMP bit in the [Section 12.1.1: SYSCFG configuration register 1 \(SYSCFG_CFGR1\) on page 248](#).

2. Available in STM32F303xD/E only.

Table 83. STM32F303x6/8 and STM32F328x8 vector table

Position	Priority	Type of priority	Acronym	Description	Address
-	-	-	-	Reserved	0x0000 0000
-	-3	Fixed	Reset	Reset	0x0000 0004
-	-2	Fixed	NMI	Non maskable interrupt. The RCC Clock Security System (CSS) is linked to the NMI vector.	0x0000 0008
-	-1	Fixed	HardFault	All class of fault	0x0000 000C
-	0	Settable	MemManage	Memory management	0x0000 0010
-	1	Settable	BusFault	Pre-fetch fault, memory access fault	0x0000 0014
-	2	Settable	UsageFault	Undefined instruction or illegal state	0x0000 0018
-	-	-	-	Reserved	0x0000 001C - 0x0000 0028
-	3	Settable	SVCall	System service call via SWI instruction	0x0000 002C
-	5	Settable	PendSV	Pendable request for system service	0x0000 0038
-	6	Settable	SysTick	System tick timer	0x0000 003C
0	7	Settable	WWDG	Window Watchdog interrupt	0x0000 0040
1	8	Settable	PVD	PVD through EXTI line 16 detection interrupt	0x0000 0044
2	9	Settable	TAMPER_STAMP	Tamper and TimeStamp interrupts through the EXTI line 19	0x0000 0048
3	10	Settable	RTC_WKUP	RTC wakeup timer interrupts through the EXTI line 20	0x0000 004C
4	11	Settable	FLASH	Flash global interrupt	0x0000 0050
5	12	Settable	RCC	RCC global interrupt	0x0000 0054
6	13	Settable	EXTI0	EXTI Line0 interrupt	0x0000 0058
7	14	Settable	EXTI1	EXTI Line1 interrupt	0x0000 005C
8	15	Settable	EXTI2_TS	EXTI Line2 and Touch sensing interrupts	0x0000 0060
9	16	Settable	EXTI3	EXTI Line3	0x0000 0064
10	17	Settable	EXTI4	EXTI Line4	0x0000 0068
11	18	Settable	DMA1_Channel1	DMA1 channel 1 interrupt	0x0000 006C
12	19	Settable	DMA1_Channel2	DMA1 channel 2 interrupt	0x0000 0070
13	20	Settable	DMA1_Channel3	DMA1 channel 3 interrupt	0x0000 0074
14	21	Settable	DMA1_Channel4	DMA1 channel 4 interrupt	0x0000 0078
15	22	Settable	DMA1_Channel5	DMA1 channel 5 interrupt	0x0000 007C
16	23	Settable	DMA1_Channel6	DMA1 channel 6 interrupt	0x0000 0080
17	24	Settable	DMA1_Channel7	DMA1 channel 7 interrupt	0x0000 0084

Table 83. STM32F303x6/8 and STM32F328x8 vector table (continued)

Position	Priority	Type of priority	Acronym	Description	Address
18	25	Settable	ADC1_2	ADC1 and ADC2 global interrupt	0x0000 0088
19	26	Settable	CAN_TX	CAN_TX interrupts	0x0000 008C
20	27	Settable	CAN_RX0	CAN_RX0 interrupts	0x0000 0090
21	28	Settable	CAN_RX1	CAN_RX1 interrupt	0x0000 0094
22	29	Settable	CAN_SCE	CAN_SCE interrupt	0x0000 0098
23	30	Settable	EXTI9_5	EXTI Line[9:5] interrupts	0x0000 009C
24	31	Settable	TIM1_BRK/TIM15	TIM1 break/TIM15 global interrupts	0x0000 00A0
25	32	Settable	TIM1_UP/TIM16	TIM1 update/TIM16 global interrupts	0x0000 00A4
26	33	Settable	TIM1_TRG_COM /TIM17	TIM1 trigger and commutation/TIM17 interrupts	0x0000 00A8
27	34	Settable	TIM1_CC	TIM1 capture compare interrupt	0x0000 00AC
28	35	Settable	TIM2	TIM2 global interrupt	0x0000 00B0
29	36	settable	TIM3	TIM3 global interrupt	0x0000 00B4
30	37	-	Reserved		0x0000 00B8
31	38	Settable	I2C1_EV	I2C1 event interrupt & EXTI Line23 interrupt	0x0000 00BC
32	39	Settable	I2C1_ER	I2C1 error interrupt	0x0000 00C0
33	40	-	Reserved		0x0000 00C4
34	41	-	Reserved		0x0000 00C8
35	42	-	SPI1	SPI1 global interrupt	0x0000 00CC
36	43	-	Reserved		0x0000 00D0
37	44	Settable	USART1	USART1 global interrupt & EXTI Line 25	0x0000 00D4
38	45	Settable	USART2	USART2 global interrupt & EXTI Line 26	0x0000 00D8
39	46	Settable	USART3	USART3 global interrupt & EXTI Line 28	0x0000 00DC
40	47	Settable	EXTI15_10	EXTI Line[15:10] interrupts	0x0000 00E0
41	48	Settable	RTC_Alarm	RTC alarm interrupt	0x0000 00E4
42	49	-	Reserved		0x0000 00E8
43	50	-	Reserved		0x0000 00EC
44	51	-	Reserved		0x0000 00F0
45	52	-	Reserved		0x0000 00F4
46	53	-	Reserved		0x0000 00F8
47	54	-	Reserved		0x0000 00FC
48	55	-	Reserved		0x0000 0100
49	56	-	Reserved		0x0000 0104

Table 83. STM32F303x6/8 and STM32F328x8 vector table (continued)

Position	Priority	Type of priority	Acronym	Description	Address
50	57	-	Reserved		0x0000 0108
51	58	-	Reserved		0x0000 010C
52	59	-	Reserved		0x0000 0110
53	60	-	Reserved		0x0000 0114
54	61	Settable	TIM6_DAC1	TIM6 global and DAC1 underrun interrupts	0x0000 0118
55	62	settable	TIM7_DAC2	TIM7 global and DAC2 underrun interrupt	0x0000 011C
56	63	-	Reserved		0x0000 0120
57	64	-	Reserved		0x0000 0124
58	65	-	Reserved		0x0000 0128
59	66	-	Reserved		0x0000 012C
60	67	-	Reserved		0x0000 0130
61	68	-	Reserved		0x0000 0134
62	69	-	Reserved		0x0000 0138
63	70	-	Reserved		0x0000 013C
64	71	Settable	COMP2	COMP2 interrupt combined with EXTI Lines 22 interrupt.	0x0000 0140
65	72	Settable	COMP4_6	COMP4 & COMP6 interrupts combined with EXTI Lines 30 and 32 interrupts respectively.	0x0000 0144
66	73	-	Reserved		0x0000 0148
67	74	-	Reserved		0x0000 014C
68	75	-	Reserved		0x0000 0150
69	76	-	Reserved		0x0000 0154
70	77	-	Reserved		0x0000 0158
71	78	-	Reserved		0x0000 015C
72	79	-	Reserved		0x0000 0160
73	80	-	Reserved		0x0000 0164
74	81	-	Reserved		0x0000 0168
75	82	-	Reserved		0x0000 016C
76	83	-	Reserved		0x0000 0170
77	84	-	Reserved		0x0000 0174
78	85	-	Reserved		0x0000 0178
79	86	-	Reserved		0x0000 017C

Table 83. STM32F303x6/8 and STM32F328x8 vector table (continued)

Position	Priority	Type of priority	Acronym	Description	Address
80	87	-	Reserved		0x0000 0180
81	88	Settable	FPU	Floating point interrupt	0x0000 0184

14.2 Extended interrupts and events controller (EXTI)

The extended interrupts and events controller (EXTI) manages the external and internal asynchronous events/interrupts and generates the event request to the CPU/Interrupt Controller and a wake-up request to the Power Manager.

The EXTI allows the management of up to 36 external/internal event line (28 external event lines and 8 internal event lines).

The active edge of each external interrupt line can be chosen independently, whilst for internal interrupt the active edge is always the rising one. An interrupt could be left pending: in case of an external one, a status register is instantiated and indicates the source of the interrupt; an event is always a simple pulse and it is used for triggering the core wake-up. For internal interrupts, the pending status is assured by the generating peripheral, so no need for a specific flag. Each input line can be masked independently for interrupt or event generation, in addition, the internal lines are sampled only in STOP mode. This controller allows also to emulate the (only) external events by software, multiplexed with the corresponding hardware event line, by writing to a dedicated register.

14.2.1 Main features

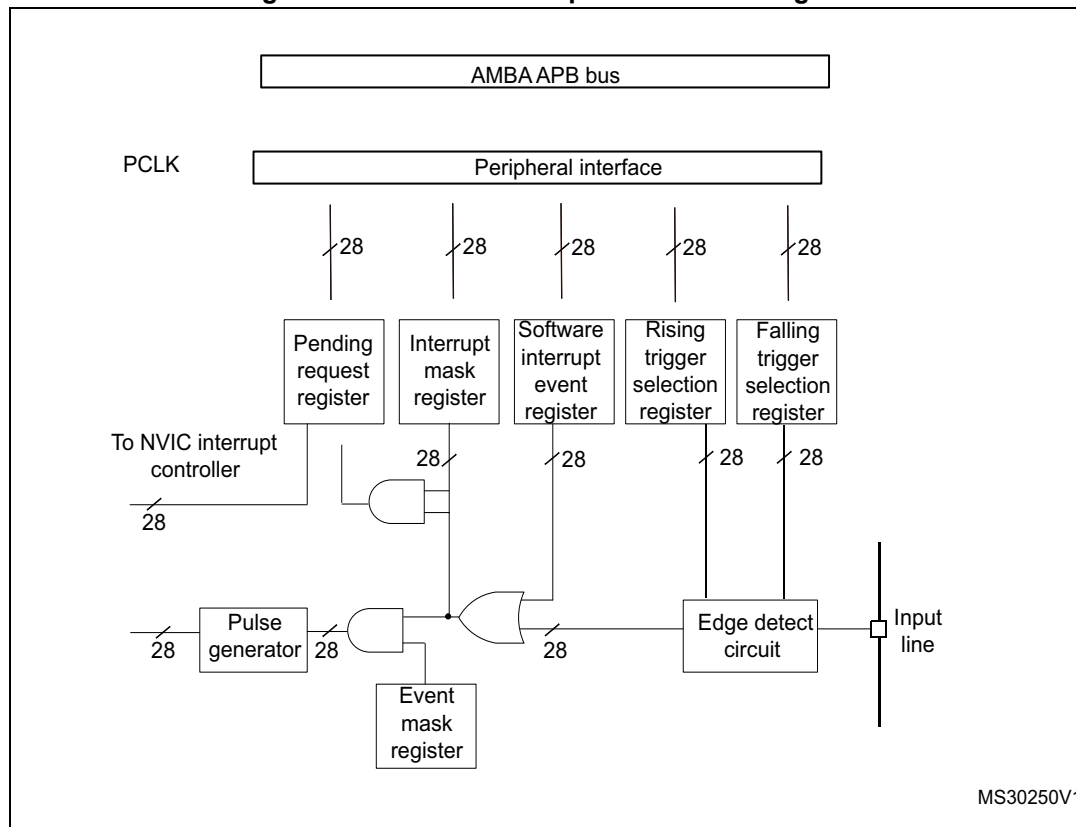
The EXTI main features are the following:

- support generation of up to 36 event/interrupt requests
- Independent configuration of each line as an external or an internal event requests
- Independent mask on each event/interrupt line
- Automatic disable of internal lines when system is not in STOP mode
- Independent trigger for external event/interrupt line
- Dedicated status bit for external interrupt line
- Emulation for all the external event requests.

14.2.2 Block diagram

The extended interrupt/event block diagram is shown in the following figure.

Figure 50. External interrupt/event block diagram



14.2.3 Wake-up event management

STM32F3xx devices are able to handle external or internal events to wake up the core (WFE). The wake-up event can be generated either by:

- enabling an interrupt in the peripheral control register but not in the NVIC, and enabling the SEVONPEND bit in the Cortex-M4 System Control register. When the MCU resumes from WFE, the EXTI peripheral interrupt pending bit and the peripheral NVIC IRQ channel pending bit (in the NVIC interrupt clear pending register) have to be cleared.
- or by configuring an external or internal EXTI line in event mode. When the CPU resumes from WFE, it is not necessary to clear the peripheral interrupt pending bit or the NVIC IRQ channel pending bit as the pending bit corresponding to the event line is not set.

14.2.4 Asynchronous Internal Interrupts

Some communication peripherals (UART, I2C) are able to generate events when the system is in run mode and also when the system is in stop mode allowing to wake up the system from stop mode.

To accomplish this, the peripheral is asked to generate both a synchronized (to the system clock, for example, APB clock) and an asynchronous version of the event.

14.2.5 Functional description

For the external interrupt lines, to generate the interrupt, the interrupt line should be configured and enabled. This is done by programming the two trigger registers with the desired edge detection and by enabling the interrupt request by writing a '1' to the corresponding bit in the interrupt mask register. When the selected edge occurs on the external interrupt line, an interrupt request is generated. The pending bit corresponding to the interrupt line is also set. This request is reset by writing a 1 in the pending register.

For the internal interrupt lines, the active edge is always the rising edge. The interrupt is enabled by default in the interrupt mask register and there is no corresponding pending bit in the pending register.

To generate the event, the event line should be configured and enabled. This is done by programming the two trigger registers with the desired edge detection and by enabling the event request by writing a '1' to the corresponding bit in the event mask register. When the selected edge occurs on the event line, an event pulse is generated. The pending bit corresponding to the event line is not set.

For the external lines, an interrupt/event request can also be generated by software by writing a 1 in the software interrupt/event register.

Note: The interrupts or events associated to the internal lines can be triggered only when the system is in STOP mode. If the system is still running, no interrupt/event is generated.

Hardware interrupt selection

To configure a line as interrupt source, use the following procedure:

- Configure the corresponding mask bit in the EXTI_IMR register.
- Configure the Trigger Selection bits of the Interrupt line (EXTI_RTSR and EXTI_FTSR)
- Configure the enable and mask bits that control the NVIC IRQ channel mapped to the EXTI so that an interrupt coming from one of the EXTI lines can be correctly acknowledged.

Hardware event selection

To configure a line as event source, use the following procedure:

- Configure the corresponding mask bit in the EXTI_EMR register.
- Configure the Trigger Selection bits of the Event line (EXTI_RTSR and EXTI_FTSR)

Software interrupt/event selection

Any of the external lines can be configured as software interrupt/event lines. The following is the procedure to generate a software interrupt.

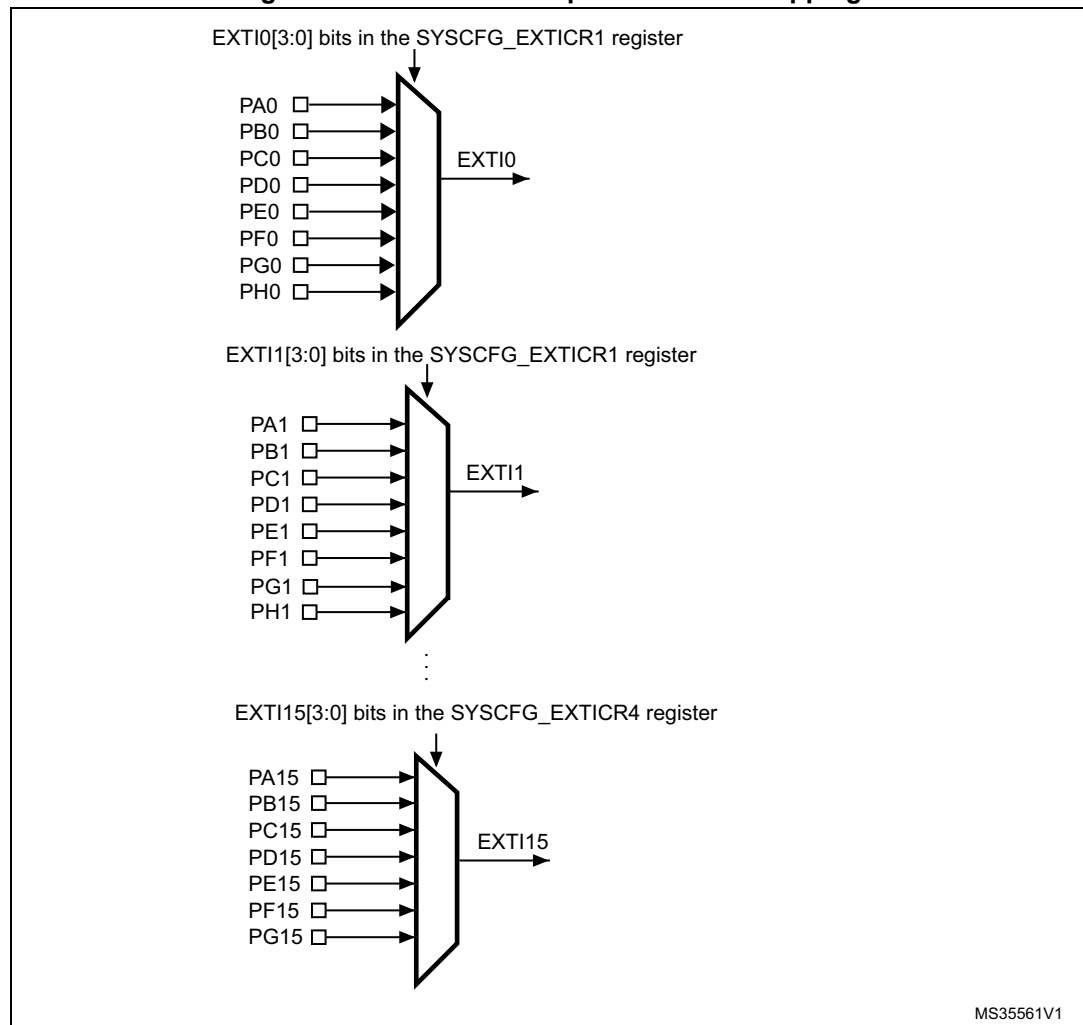
- Configure the corresponding mask bit (EXTI_IMR, EXTI_EMR)
- Set the required bit of the software interrupt register (EXTI_SWIER)

14.2.6 External and internal interrupt/event line mapping

36 interrupt/event lines are available: 8 lines are internal (including the reserved ones); the remaining 28 lines are external.

The GPIOs are connected to the 16 external interrupt/event lines in the following manner:

Figure 51. External interrupt/event GPIO mapping



The remaining lines are connected as follows:

- EXTI line 16 is connected to the PVD output
- EXTI line 17 is connected to the RTC Alarm event
- EXTI line 18 is connected to USB Device FS wake-up event (STM32F303xB/C/D/E, STM32F358xC and STM32F398xE devices)
- EXTI line 19 is connected to RTC tamper and timestamps
- EXTI line 20 is connected to RTC wake-up timer
- EXTI line 21 is connected to Comparator 1 output (STM32F303xB/C/D/E, STM32F358xC and STM32F398xE devices)
- EXTI line 22 is connected to Comparator 2 output
- EXTI line 23 is connected to I2C1 wake-up
- EXTI line 24 is connected to I2C2 wake-up (STM32F303xB/C/D/E, STM32F358xC and STM32F398xE devices)
- EXTI line 25 is connected to USART1 wake-up
- EXTI line 26 is connected to USART2 wake-up (STM32F303xB/C/D/E, STM32F358xC and STM32F398xE devices)
- EXTI line 27 is connected to I2C3 wake-up (STM32F303xD/E and STM32F398xE devices)
- EXTI line 28 is connected to USART3 wake-up (STM32F303xB/C/D/E, STM32F358xC and STM32F398xE devices)
- EXTI line 29 is connected to Comparator 3 output (STM32F303xB/C/D/E, STM32F358xC and STM32F398xE devices)
- EXTI line 30 is connected to Comparator 4 output
- EXTI line 31 is connected to Comparator 5 output (STM32F303xB/C/D/E, STM32F358xC and STM32F398xE devices)
- EXTI line 32 is connected to Comparator 6 output
- EXTI line 33 is connected to Comparator 7 output (STM32F303xB/C/D/E, STM32F358xC and STM32F398xE devices)
- EXTI line 34 is connected to UART4 wake-up (STM32F303xB/C/D/E, STM32F358xC and STM32F398xE devices)
- EXTI line 35 is connected to UART5 wake-up (STM32F303xB/C/D/E, STM32F358xC and STM32F398xE devices)

Note: EXTI lines 23, 24, 25, 26, 27, 28, 34 and 35 are internal.

14.3 EXTI registers

Refer to [Section 2.2](#) for a list of abbreviations used in register descriptions.

The peripheral registers have to be accessed by words (32-bit).

14.3.1 Interrupt mask register (EXTI_IMR1)

Address offset: 0x00

Reset value: 0x1F80 0000 (Refer to the note below)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
MR31	MR30	MR29	MR28	MR27	MR26	MR25	MR24	MR23	MR22	MR21	MR20	MR19	MR18	MR17	MR16
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
MR15	MR14	MR13	MR12	MR11	MR10	MR9	MR8	MR7	MR6	MR5	MR4	MR3	MR2	MR1	MR0
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw

Bits 31:0 **MRx**: Interrupt Mask on external/internal line x

0: Interrupt request from Line x is masked

1: Interrupt request from Line x is not masked

Note: The reset value for the internal lines (23, 24, 25, 26, 27 and 28) is set to '1' to enable the interrupt by default.

14.3.2 Event mask register (EXTI_EMR1)

Address offset: 0x04

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
MR31	MR30	MR29	MR28	MR27	MR26	MR25	MR24	MR23	MR22	MR21	MR20	MR19	MR18	MR17	MR16
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
MR15	MR14	MR13	MR12	MR11	MR10	MR9	MR8	MR7	MR6	MR5	MR4	MR3	MR2	MR1	MR0
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw

Bits 31:0 **MRx**: Event Mask on external/internal line x

0: Event request from Line x is masked

1: Event request from Line x is not masked

14.3.3 Rising trigger selection register (EXTI_RTSTR1)

Address offset: 0x08

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
TR31	TR30	TR29	Res.	Res.	Res.	Res.	Res.	Res.	TR22	TR21	TR20	TR19	TR18	TR17	TR16
rW	rW	rW							rW	rW	rW	rW	rW	rW	rW
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
TR15	TR14	TR13	TR12	TR11	TR10	TR9	TR8	TR7	TR6	TR5	TR4	TR3	TR2	TR1	TR0
rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW

Bits 31:29 **TRx**: Rising trigger event configuration bit of line x (x = 31 to 29)

0: Rising trigger disabled (for Event and Interrupt) for input line

1: Rising trigger enabled (for Event and Interrupt) for input line.

Bits 28:23 Reserved, must be kept at reset value.

Bits 22:0 **TRx**: Rising trigger event configuration bit of line x (x = 22 to 0)

0: Rising trigger disabled (for Event and Interrupt) for input line

1: Rising trigger enabled (for Event and Interrupt) for input line.

Note: *The external wake-up lines are edge-triggered. No glitches must be generated on these lines. If a rising edge on an external interrupt line occurs during a write operation in the EXTI_RTSTR register, the pending bit is not set.*

Rising and falling edge triggers can be set for the same interrupt line. In this case, both generate a trigger condition.

14.3.4 Falling trigger selection register (EXTI_FTSR1)

Address offset: 0x0C

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
TR31	TR30	TR29	Res.	Res.	Res.	Res.	Res.	Res.	TR22	TR21	TR20	TR19	TR18	TR17	TR16
rW	rW	rW							rW	rW	rW	rW	rW	rW	rW
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
TR15	TR14	TR13	TR12	TR11	TR10	TR9	TR8	TR7	TR6	TR5	TR4	TR3	TR2	TR1	TR0
rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW

Bits 31:29 **TRx**: Falling trigger event configuration bit of line x (x = 31 to 29)

0: Falling trigger disabled (for Event and Interrupt) for input line

1: Falling trigger enabled (for Event and Interrupt) for input line.

Bits 28:23 Reserved, must be kept at reset value.

Bits 22:0 **TRx**: Falling trigger event configuration bit of line x (x = 22 to 0)

0: Falling trigger disabled (for Event and Interrupt) for input line

1: Falling trigger enabled (for Event and Interrupt) for input line.

Note: *The external wake-up lines are edge-triggered. No glitches must be generated on these lines. If a falling edge on an external interrupt line occurs during a write operation to the EXTI_FTSR register, the pending bit is not set.*

Rising and falling edge triggers can be set for the same interrupt line. In this case, both generate a trigger condition.

14.3.5 Software interrupt event register (EXTI_SWIER1)

Address offset: 0x10

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
SWIER 31	SWIER 30	SWIER 29	Res.	Res.	Res.	Res.	Res.	Res.	SWIER 22	SWIER 21	SWIER 20	SWIER 19	SWIER 18	SWIER 17	SWIER 16
rw	rw	rw							rw	rw	rw	rw	rw	rw	rw
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
SWIER 15	SWIER 14	SWIER 13	SWIER 12	SWIER 11	SWIER 10	SWIER 9	SWIER 8	SWIER 7	SWIER 6	SWIER 5	SWIER 4	SWIER 3	SWIER 2	SWIER 1	SWIER 0
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw

Bits 31: 29 **SWIERx**: Software interrupt on line x (x = 31 to 29)

If the interrupt is enabled on this line in the EXTI_IMR, writing a '1' to this bit when it is at '0' sets the corresponding pending bit in EXTI_PR resulting in an interrupt request generation.

This bit is cleared by clearing the corresponding bit in the EXTI_PR register (by writing a '1' into the bit).

Bits 22:0 **SWIERx**: Software interrupt on line x (x = 22 to 0)

If the interrupt is enabled on this line in the EXTI_IMR, writing a '1' to this bit when it is at '0' sets the corresponding pending bit in EXTI_PR resulting in an interrupt request generation.

This bit is cleared by clearing the corresponding bit of EXTI_PR (by writing a '1' into the bit).

14.3.6 Pending register (EXTI_PR1)

Address offset: 0x14

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
PR31	PR30	PR29	Res.	Res.	Res.	Res.	Res.	Res.	PR22	PR21	PR20	PR19	PR18	PR17	PR16
rc_w1	rc_w1	rc_w1							rc_w1	rc_w1	rc_w1	rc_w1	rc_w1	rc_w1	rc_w1
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
PR15	PR14	PR13	PR12	PR11	PR10	PR9	PR8	PR7	PR6	PR5	PR4	PR3	PR2	PR1	PR0
rc_w1	rc_w1	rc_w1	rc_w1	rc_w1	rc_w1	rc_w1	rc_w1	rc_w1	rc_w1	rc_w1	rc_w1	rc_w1	rc_w1	rc_w1	rc_w1

Bits 31:29 **PRx**: Pending bit on line x (x = 31 to 29)

0: No trigger request occurred

1: Selected trigger request occurred

This bit is set when the selected edge event arrives on the external interrupt line.

This bit is cleared by writing a '1' to the bit.

Bits 28:23 Reserved, must be kept at reset value.

Bits 22:0 **PRx**: Pending bit on line x (x = 22 to 0)

0: No trigger request occurred

1: Selected trigger request occurred

This bit is set when the selected edge event arrives on the external interrupt line.

This bit is cleared by writing a '1' to the bit.

14.3.7 Interrupt mask register (EXTI_IMR2)

Address offset: 0x20

Reset value: 0xFFFF FFFC (See note below)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	MR35	MR34	MR33	MR32
												rw	rw	rw	rw

Bits 31:4 Reserved, must be kept at reset value

Bits 3:0 **MRx**: Interrupt mask on external/internal line x, x = 32..35

0: Interrupt request from Line x is masked

1: Interrupt request from Line x is not masked

Note: The reset value for the internal lines (34 and 35) and reserved lines is set to '1'.

14.3.8 Event mask register (EXTI_EMR2)

Address offset: 0x24

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	MR35	MR34	MR33	MR32
												rw	rw	rw	rw

Bits 31:4 Reserved, must be kept at reset value

Bits 3:0 **MRx**: Event mask on external/internal line x, x = 32..35

0: Event request from Line x is masked

1: Event request from Line x is not masked

14.3.9 Rising trigger selection register (EXTI_RTSTR2)

Address offset: 0x28

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	TR33	TR32
														rw	rw

Bits 31:2 Reserved, must be kept at reset value.

Bits 1:0 **TRx**: Rising trigger event configuration bit of line x (x = 32, 33)

0: Rising trigger disabled (for Event and Interrupt) for input line

1: Rising trigger enabled (for Event and Interrupt) for input line.

Note: *The external wake-up lines are edge-triggered. No glitches must be generated on these lines. If a rising edge on an external interrupt line occurs during a write operation to the EXTI_RTSTR register, the pending bit is not set.*

Rising and falling edge triggers can be set for the same interrupt line. In this case, both generate a trigger condition.

14.3.10 Falling trigger selection register (EXTI_FTSR2)

Address offset: 0x2C

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	TR33	TR32
														rw	rw

Bits 31:2 Reserved, must be kept at reset value.

Bits 1:0 **TRx**: Falling trigger event configuration bit of line x (x = 32, 33)

0: Falling trigger disabled (for Event and Interrupt) for input line

1: Falling trigger enabled (for Event and Interrupt) for input line.

Note: The external wake-up lines are edge-triggered. No glitches must be generated on these lines. If a falling edge on an external interrupt line occurs during a write operation to the EXTI_FTSR register, the pending bit is not set.

Rising and falling edge triggers can be set for the same interrupt line. In this case, both generate a trigger condition.

14.3.11 Software interrupt event register (EXTI_SWIER2)

Address offset: 0x30

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	SWIER 33	SWIER 32
														rw	rw

Bits 31:2 Reserved, must be kept at reset value.

Bits 1:0 **SWIERx**: Software interrupt on line x (x = 32, 33)

If the interrupt is enabled on this line in the EXTI_IMR, writing a '1' to this bit when it is at '0' sets the corresponding pending bit in EXTI_PR resulting in an interrupt request generation.

This bit is cleared by clearing the corresponding bit of EXTI_PR (by writing a '1' to the bit).

14.3.12 Pending register (EXTI_PR2)

Address offset: 0x34

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	PR33	PR32
														rc_w1	rc_w1

Bits 31:2 Reserved, must be kept at reset value.

Bits 1:0 **PRx**: Pending bit on line x (x = 32,33)

0: No trigger request occurred

1: Selected trigger request occurred

This bit is set when the selected edge event arrives on the external interrupt line.

This bit is cleared by writing a '1' into the bit.

14.3.13 EXTI register map

Table 84. External interrupt/event controller register map and reset values

Offset	Register	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0		
0x00	EXTI_IMR1	MR[31:0]																																	
	Reset value	0	0	0	1	1	1	1	1	1	0		0	0		0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0		
0x04	EXTI_EMR1	MR[31:0]																																	
	Reset value	0	0	0	0		0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0		
0x08	EXTI_RTISR1	TR[31:29]			Res.	Res.	Res.	Res.	Res.	Res.	Res.	TR[22:0]																							
	Reset value	0	0	0								0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0		
0x0C	EXTI_FTSR1	TR[31:29]			Res.	Res.	Res.	Res.	Res.	Res.	Res.	TR[22:0]																							
	Reset value	0	0	0								0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0		
0x10	EXTI_SWIER1	SWIER [31:29]			Res.	Res.	Res.	Res.	Res.	Res.	Res.	SWIER[22:0]																							
	Reset value	0	0	0								0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0		
0x14	EXTI_PR1	PR [31:29]			Res.	Res.	Res.	Res.	Res.	Res.	PR[22:0]																								
	Reset value	0	0	0								0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0		
0x20	EXTI_IMR2	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	MR35	MR34	MR33	MR32		
	Reset value																													1	1	0	0		
0x24	EXTI_EMR2	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	MR35	MR34	MR33	MR32	
	Reset value																													0	0	0	0		
0x28	EXTI_RTISR2	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	TR33	TR32	TR31	
	Reset value																															0	0	0	
0x2C	EXTI_FTSR2	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	TR33	TR32	TR31
	Reset value																															0	0	0	
0x30	EXTI_SWIER2	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	SWIER33	SWIER32	SWIER31
	Reset value																															0	0	0	
0x34	EXTI_PR2	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	PR33	PR32	PR31
	Reset value																															0	0	0	

Refer to [Section 3.2 on page 53](#) for the register boundary addresses.